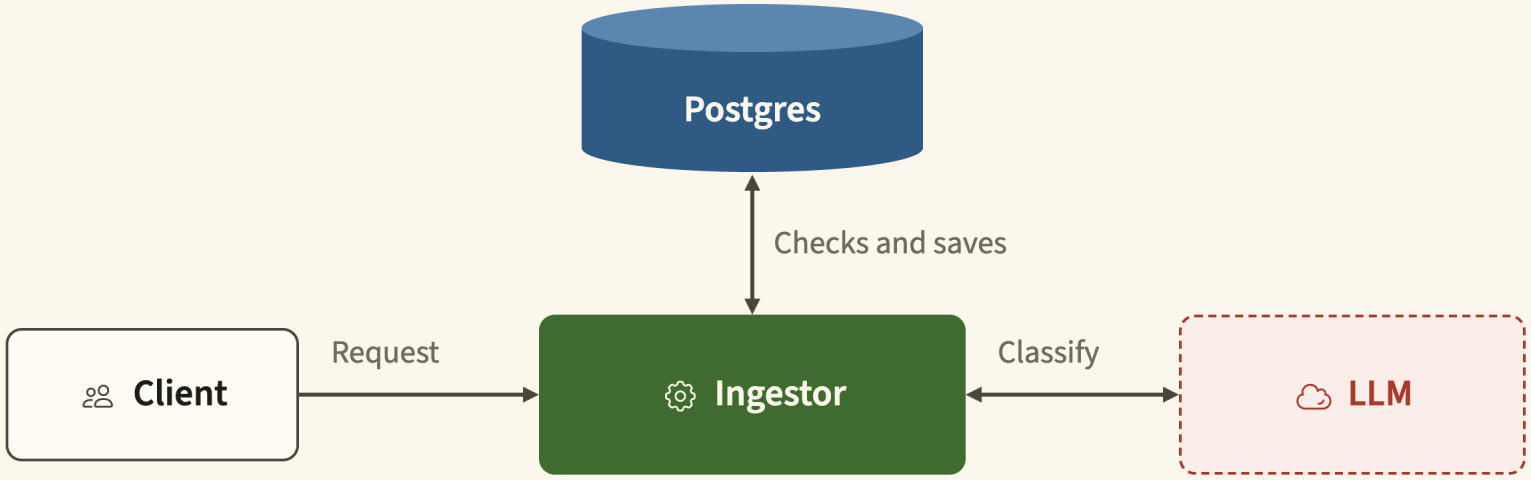
ENGINEERING TAKE-HOME · NOV 2026

Support ticket *classifier*

From the naive design to a correct one

One service does *everything*

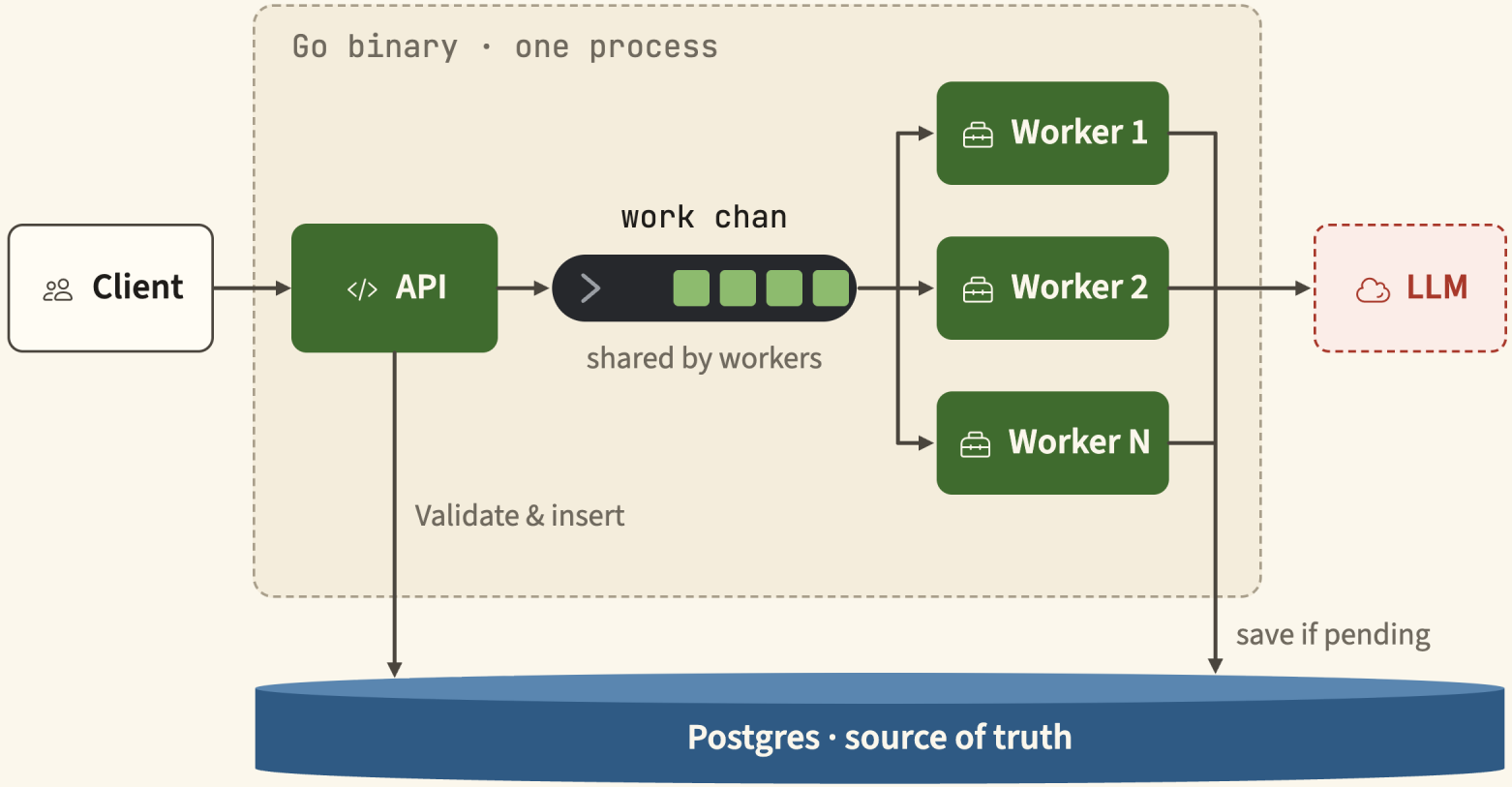


What breaks

1. No separation of concerns
2. We cannot scale
3. Read & Write commands go to the same path

One channel, *many workers*

Note : Concepts remain same, tools change in other words Channel can be a queue broker



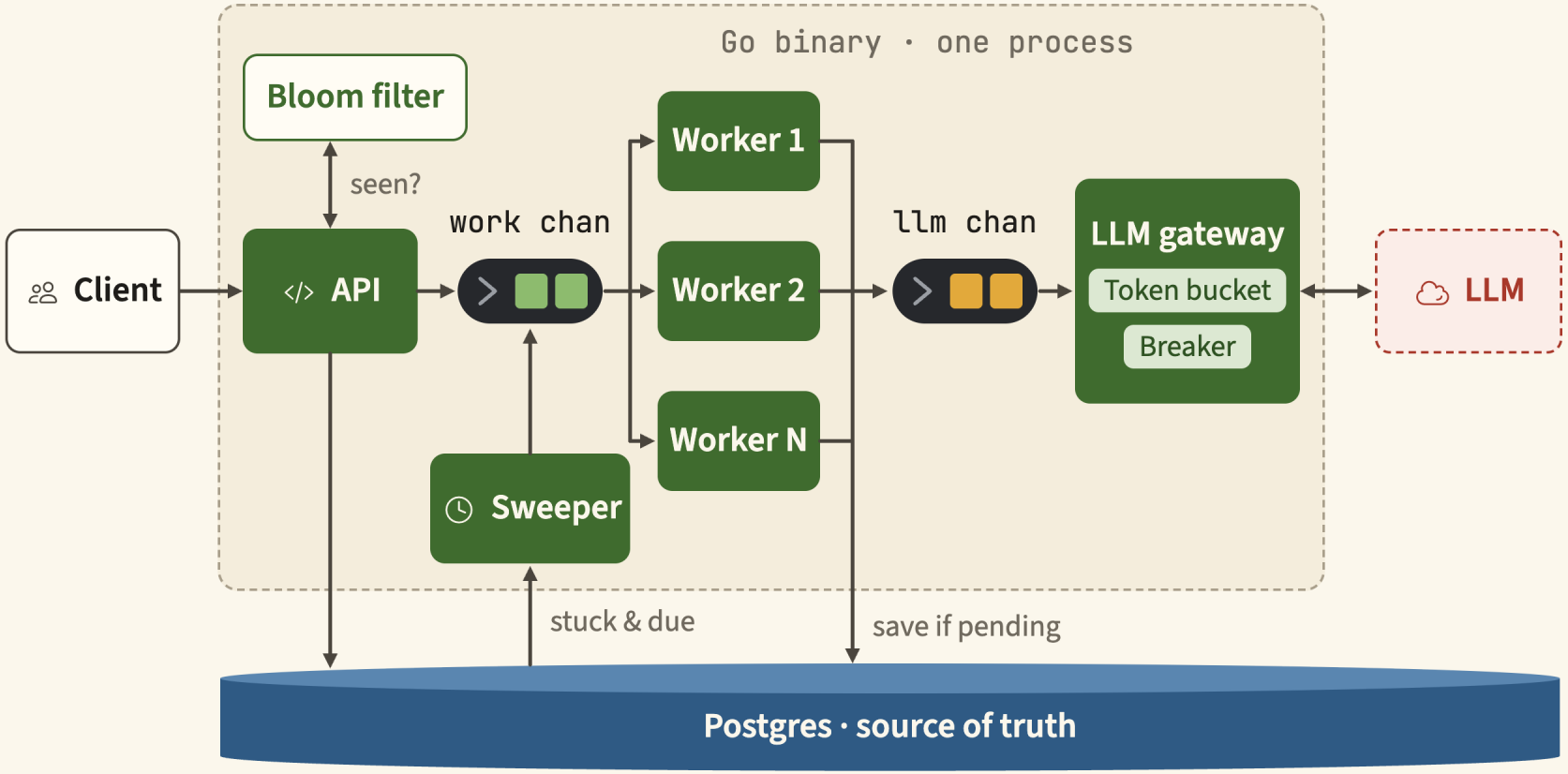
Benefits

- Async introduced
- stateless, then we can scale
- Better seperation of concerns
- Backpressure

Issues

- No crash recovery (negotiable)
- No smart awarness e.g. retires and spikes handling
- Heavy loading the Database (espically validating the ticket)

Same design, now *resilient*



Benefits

- Sweeper recovers stuck tickets
- Gateway: token bucket and breaker
- Bloom filter for duplicates

Cherry on top the cake

- Inbound rate limiting
- Optional webhooks
- Why no caches
- CQRS